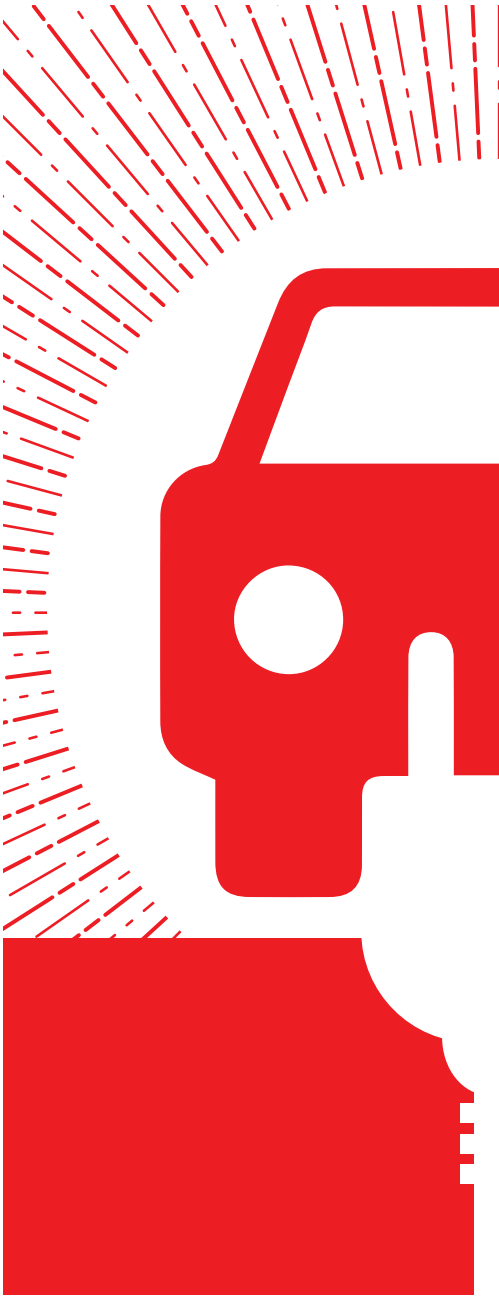
(BaaS) offering infrastructure, front-ended by existing distribution networks, come up in the top 10 urban agglomerations, thus enabling smaller—and therefore cheaper—lithium-ion batteries. Simultaneously, leading-edge players could incorporate newer technologies that do away with rare-earth magnets in their motors and pass on the direct cost reduction benefits to consumers—but, equally important, ensure greater propulsion system efficiencies, in turn enabling greater range with a given battery capacity. Competencies in embedded software will also rapidly reduce costly import dependence on controllers and battery management systems, as production capabilities in areas like mobile phone manufacture spread across sectors.

Thus, with existing tax structures, a high-speed E2W could well become available to consumers—to own—at just Rs 100 a day for three years, including a monthly 'packaged-energy' service, a little more than the price of a litre of petrol! Combined with the ability of 'disruptors' to harness technology to rapidly broaden and deepen their product range, the market will witness a proliferation of product offerings to meet a variety of use cases, covering the entire range of 2W from mopeds to high-engine-capacity (I didn't say high-performance) motorcycles. With all the attendant benefits of vastly superior performance, modern consumer-machine interface, and close to zero maintenance spends, the economics of the E2W could be so alluring so as to create a tsunami of consumers shifting away from ICE products.

Over the past couple of years, as the industry's focus has shifted from metal-bashing competencies to deep technology capabilities, the 'disruptors' have changed the paradigms of production and distribution efficiencies and started a move into a direct relationship with consumers—a move with interesting cost efficiencies that legacy players could well emulate in part. Are we then going to witness a dramatic change in industry structure? Quite possibly. The technology-focused new entrants could well account for over 40 per cent of the total market volumes of E2W by the end of 2023. One legacy player that embraced the EV revolution ahead of its peers and built significant technology capabilities to layer on top of its highly acclaimed product quality, and a global 2W giant well-entrenched in India appear best set to gain significant consumer interest. And the rest will probably have to remain content with a highly fragmented residual pie. Mergers or mortality beckon, and a wave of consolidation is to be expected. And don't be surprised to see a dominance of Indian brands.

> THE ECONOMICS OF ELECTRIC TWO-WHEELERS **COULD CREATE A TSUNAMI OF CONSUMERS SHIFTING** AWAY FROM TRADITIONAL ENGINES

To this boiling pot, could we add a dash of spice? Imagine the effect of a component manufacturer entering the market with a retro-fitment kit to convert existing ICE 2W into either fully electric or strong hybrids. The potential market could well be over 150 million users. And the demonstration effect would be so strong and widespread that the entire new vehicle market could turn electric well before 2030. But that requires a brand-building mindset—not one that is content with arithmetic growth by virtue of latching on to the coat-tails of original equipment manufacturers (OEMs) and living with a mere margin focus. Perhaps, we will see a technology-focused new entrant in the component business too. But if it does happen, imagine the immediate impact on urban pollution.



Meanwhile, the 3W market has embraced electrification wholeheartedly. E3Ws and E-rickshaws combined already account for half the 3W registered in 2022. While volumes of the ICE variants have fallen 50 per